Tight Mortality Bounds for Image-Bounded Nondeterministic Automata

ZIJIAN ZENG, UCSI University, Malaysia

A nondeterministic finite automaton is k-image-bounded if every word maps every individual state to at most k states. A word is mortal if it maps the entire state set to the empty set. Kiefer and Ryzhikov showed that an incomplete n-state k-image-bounded automaton has a mortal word of length at most n^{k+1} and asked whether this bound is tight.

We resolve the fixed-parameter tightness question: for every fixed integer $k \geq 2$, the maximum shortest mortal-word length is $\Theta_k(n^{k+1})$. The lower bound holds for every sufficiently large n, even for binary strongly connected automata. The construction combines a linear-size queue with a packet distributed among k cyclic clocks. The clocks realize a long least-common-multiple delay using only the sum of their lengths in states. A global image invariant preserves the bound k under every word, and an exact potential forces a linear number of disjoint full-delay events. We give explicit constants, a constructive mortality argument, and a self-contained proof of the matching known upper bound.

Additional Key Words and Phrases: nondeterministic finite automata, image-bounded automata, mortality, extremal word length

Author's contact: Zijian Zeng, UCSI University, 1002266693@ucsiuniversity.edu.my.

1 The question and the result

An automaton can admit a word that eliminates every possible computation, yet require a long word to do so. Bounding the number of endpoints reachable from each state gives a natural hierarchy between deterministic automata and unrestricted nondeterministic automata. The endpoint bound must hold after every word: a bound on the number of transitions carrying one letter does not suffice.

Formally, an NFA is a finite labelled relational semiautomaton $\mathcal{A} = (Q, \Sigma, \Delta)$, with no designated initial or final states. For $X \subseteq Q$ and $w \in \Sigma^*$, write $X \cdot w$ for the union of the endpoints of all w-labelled paths starting in X; abbreviate $\{q\} \cdot w$ to $q \cdot w$. Missing transitions are allowed. The automaton is *k-image-bounded* if

$$|q \cdot w| \le k \qquad (q \in Q,\ w \in \Sigma^*). \tag{1}$$

A word w is *mortal* if $Q \cdot w = \varnothing$. We call the automaton *incomplete* when a mortal word exists, and denote the minimum length of such a word by $\mu(\mathcal{A})$. This terminology follows Kiefer and Ryzhikov [2]; incompleteness here means existence of a mortal word, rather than merely existence of a missing transition.

Let $M_k(n)$ be the maximum of $\mu(\mathcal{A})$ over incomplete n-state k-image-bounded NFAs and finite alphabets. This maximum exists: on a fixed state set there are finitely many letter relations, and duplicate relations do not affect shortest lengths. Kiefer and Ryzhikov [2, Section 9] give the bound $M_k(n) \le n^{k+1}$ and ask whether it is tight. We answer the fixed-k asymptotic interpretation of that question.

Theorem 1. *For each integer $k \ge 2$, set*

$$A_k = 1 + \frac{k(k+1)}{2}, \qquad N_k = 2k + 2A_k k!, \qquad c_k = \frac{k!}{(2A_k)^{k+1}}.$$

For every integer $n \ge N_k$, there exists a binary strongly connected incomplete n-state k-image-bounded NFA $\mathcal{A}$ satisfying

$$\mu(\mathcal{A}) \ge c_k n^{k+1}.$$

Consequently, for $n \ge N_k$,

$$c_k n^{k+1} \le M_k(n) \le n^{k+1},$$

and $M_k(n) = \Theta_k(n^{k+1})$ for every fixed $k \ge 2$.

The constants and threshold may depend on k; the theorem determines the exponent at every fixed level $k \ge 2$. The general subset bound $2^n - 1$ limits the joint dependence on k and n. Strong connectivity means that the underlying directed graph is strongly connected. Our lower-bound examples have this property and a binary alphabet. The quadratic order at $k = 1$ is known.

1.1 Prior mechanisms and the additional ingredient

Ryzhikov [3, Section 4] describes a binary deterministic construction with a cyclic queue and a physical delay path. Our queue action uses this known mechanism, up to indexing. His discussion also recalls the sharp arbitrary-alphabet deterministic mortality bound $n(n+1)/2$, attributed there to Rystsov. The unrestricted NFA exponential lower bounds in the same paper do not impose (1).

Repeated deterministic delays also occur in Vorel's appended-tail lemma, reproduced with proof by Ananichev [1, Section 2, Lemma 2.1]. Under its almost-permutation and divisibility hypotheses, adding a tail of h states to an r-state automaton increases its reset threshold by rh. That lemma concerns synchronization to a sink; we use it as prior context for the delay mechanism, not as a theorem about the present NFA model.

The additional ingredient is a reusable nondeterministic delay. A packet with one state in each of k cycles has period $L = \operatorname{lcm}(\ell_1, \dots, \ell_k)$, although it uses only $\sum_i \ell_i$ clock states. Pairwise coprime lengths can make L of order n^k at fixed k. The essential issue is to force a linear number of these delays without allowing

resets, queue collisions, or simultaneous forks to create a shortcut. We arrange for every failed packet to return to a single queue port, separate from the injective action on the remaining queue. The potential in Lemma 4 then measures exactly how much progress any word can make. Thus the same clocks are reused, rather than copied for each queue state. This supplies the additional factor n while preserving the global endpoint promise.

2 A queue and a packet of clocks

Fix integers $k \geq 2$, $m \geq 2$, and $\ell_i \geq 2$ for $1 \leq i \leq k$. The disjoint state sets are

$$P = \{p_0, \ldots, p_{m-1}\}, \qquad C_i = \{c_{i,0}, \ldots, c_{i,\ell_i-1}\}, \qquad Q = P \cup \bigcup_{i=1}^{k} C_i.$$

Let $S = \{c_{i,0} : 1 \leq i \leq k\}$ and $L = \operatorname{lcm}(\ell_1, \ldots, \ell_k)$. The alphabet is $\{a, b\}$. Letter a rotates each of $P, C_1, \ldots, C_k$ forward by one position:

$$p_j \cdot a = \{p_{j+1 \bmod m}\}, \qquad c_{i,j} \cdot a = \{c_{i,j+1 \bmod \ell_i}\}.$$

The other letter is defined by

$$p_0 \cdot b = S, \qquad p_1 \cdot b = \{p_{m-1}\}, \qquad p_j \cdot b = \{p_{j-1}\} \quad (2 \leq j < m), \tag{2}$$

$$c_{i,j} \cdot b = \begin{cases} \{p_0\}, & 0 \leq j < \ell_i - 1, \\ \varnothing, & j = \ell_i - 1. \end{cases} \tag{3}$$

Write f for the permutation of $P \setminus \{p_0\}$ induced by b. For $m = 2$ this permutation fixes p_1. Call $S \cdot a^t$ a *packet*; its position in every clock is determined by the same t. The state c_{i,ℓ_i-1} is the terminal position of clock i.

Proposition 2. *The automaton just defined is strongly connected and globally k-image-bounded, with least possible image bound exactly k. It is incomplete and satisfies*

$$1 + (m+1)L \;\leq\; \mu(\mathcal{A}) \;\leq\; 1 + (m+1)L + m^2. \tag{4}$$

We prove the properties separately. The clock lengths need not be coprime for Proposition 2, and no condition on $\gcd(m, L)$ is needed.

Lemma 3. *Every image of a singleton under any word is empty, a singleton, or a packet. The underlying directed graph is strongly connected.*

Proof. The family consisting of $\varnothing$, all singletons, and all packets is closed under a. By (2)–(3), the b-image of a singleton is empty, a singleton, or S. For a packet, each component either disappears under b or reaches the same state p_0. Its image is therefore empty or $\{p_0\}$. This family is closed under both letters and contains every initial singleton, so induction on the whole word proves the assertion. Each member has size at most k, and $|p_0 \cdot b| = k$ proves equality of the least admissible bound. This induction covers arbitrarily many forks.

Every queue state reaches p_0 by a power of a. Every clock state reaches its position zero by a power of a and then reaches p_0 by b, since $\ell_i \geq 2$. Conversely, from p_0 powers of a reach the entire queue; b followed by suitable powers of a has a path to any prescribed clock state. These paths prove strong connectivity. □

3 Charging every unit of progress

We examine global images immediately after occurrences of b, called *b-boundaries*. At such a boundary the image has the form

$$T \cup E_\eta, \qquad T \subseteq P, \quad \eta \in \{0,1\}, \qquad E_0 = \varnothing, \quad E_1 = S.$$

The representation is unique because queue and clock states are disjoint. For this image define

$$V(T,\eta) = |T| + \eta.$$

This counts queue states and one unit for an entire packet, rather than counting the individual states in that packet.

Lemma 4. *Every mortal word has length at least* $1 + (m+1)L$.

Proof. Letter a is a permutation of Q, and

$$Q \cdot b = P \cup S. \tag{5}$$

Indeed, p_0 supplies S, the other queue states supply $P \setminus \{p_0\}$, and the nonterminal clock states supply p_0. Thus the potential immediately after the first b is $m+1$.

Consider any interval $a^t b$, $t \geq 0$, between consecutive occurrences of b. Suppose its starting image is $T \cup E_\eta$. Put $U = T \cdot a^t$, let α be the indicator of $p_0 \in U$, and let ε be the indicator that $\eta = 1$ and at least one state in $S \cdot a^t$ is nonterminal. After this interval the queue part is

$$T' = f(U \setminus \{p_0\}) \cup \begin{cases} \{p_0\}, & \varepsilon = 1, \\ \varnothing, & \varepsilon = 0, \end{cases}$$

and the packet indicator is $\eta' = \alpha$. The displayed union is disjoint, and f is injective. Since a permutes P, we obtain the exact identity

$$V(T',\eta') = |T| - \alpha + \varepsilon + \alpha = |T| + \varepsilon, \qquad V(T,\eta) - V(T',\eta') = \eta - \varepsilon. \tag{6}$$

In particular, the representation persists after every b.

The drop is either zero or one. It equals one precisely when a packet was present and every component of that packet reaches a terminal clock position. Since its starting positions are all zero, this condition is

$$t \equiv -1 \pmod{\ell_i} \quad \text{for every } i, \qquad \text{equivalently} \qquad L \mid t+1.$$

Every interval that lowers V therefore has length $t+1 \geq L$. The identity includes the case where a killed packet is replaced by a new packet at the very same b: then $\alpha = 1$, and the potential still drops by exactly one. Consecutive b's and empty queue subsets are also covered.

A mortal word contains b, because a alone is a permutation. Delete an initial run of a's, which leaves Q unchanged, and a terminal run of a's, which cannot turn a nonempty subset into an empty one. The remaining mortal word starts and ends with b. By (5), its first b gives potential $m+1$; its last gives zero. Equation (6) forces exactly $m+1$ unit drops. The intervals between consecutive b's are disjoint, and each of these drops costs at least L letters. Including the first b proves the lower bound for the shortened word and hence for the original word. □

Lemma 5. *There is a mortal word of length at most* $1 + (m+1)L + m^2$.

Proof. Begin with b, giving potential $m+1$ and a packet. At any b-boundary with a packet, apply $a^{L-1}b$. By (6), this lowers the potential by exactly one, whatever the queue subset is.

If the packet is absent but the potential is positive, choose any active queue state p_j. Take $t \in \{0,\dots,m-1\}$ with $j + t \equiv 0 \pmod{m}$, and apply $a^t b$. This creates a packet and leaves the potential unchanged: in (6)

we have $\eta = \varepsilon = 0$ and $\alpha = 1$. This preliminary fork costs at most m letters. Now apply $a^{L-1}b$ to lower the potential by one.

After exactly $m+1$ such drops the potential is zero, so the global image is empty. The first drop needs no preliminary fork, and at most m later forks are needed. The total length is at most $1 + (m+1)L + m^2$. □

Lemmas 3–5 prove Proposition 2, including an explicit mortality procedure.

4 Obtaining every sufficiently large state count

Proof of the lower bound in Theorem 1. Fix $k \geq 2$, write $K = k!$ and $A = A_k$, and take any $n \geq N_k$. Choose

$$M = K\left\lfloor \frac{n-k}{AK} \right\rfloor, \qquad \ell_i = 1 + iM \ (1 \leq i \leq k), \qquad m = n - k - (A-1)M. \tag{7}$$

Since $AM \leq n-k$, we have $m \geq M$. The floor estimate gives

$$M \geq \frac{n-k}{A} - K \geq \frac{n}{2A},$$

where the second inequality follows from $n \geq 2k + 2AK$. Also $(n-k)/(AK) \geq 2$, so $M \geq 2K \geq 2$. Thus $m \geq 2$ and $\ell_i \geq 2$, as required. Moreover,

$$m + \sum_{i=1}^{k} \ell_i = m + k + \frac{k(k+1)}{2}M = n.$$

There is no padding by extra states, and strong connectivity is retained.

The lengths ℓ_i are pairwise coprime. To see this, suppose $i < j$ and d divides both $1 + iM$ and $1 + jM$. Then $\gcd(d, M) = 1$, while $d \mid (j-i)M$, so $d \mid j - i$. If $d > 1$, a prime divisor p of d satisfies $p \leq j - i \leq k - 1$ and therefore divides K and M. This contradicts $\gcd(d, M) = 1$. Hence $d = 1$.

It follows that

$$L = \prod_{i=1}^{k}(1 + iM) \geq KM^k.$$

Proposition 2 now yields

$$\mu(\mathcal{A}) \geq 1 + (m+1)L \geq KM^{k+1} \geq \frac{k!}{(2A_k)^{k+1}}n^{k+1}.$$

The construction is binary, strongly connected, and globally k-image-bounded by the same proposition, proving the claimed lower bound for every $n \geq N_k$. □

5 The matching upper bound

For completeness we reproduce the elementary upper argument behind the bound of Kiefer and Ryzhikov [2, Section 9]. Let $\mathcal{A}$ be any incomplete n-state k-image-bounded NFA and put

$$D = \sum_{j=1}^{\min(k,n)} \binom{n}{j}.$$

Consider any nonempty image $q \cdot u$ of an original state. A killing continuation exists, since a global mortal word kills every subset. Along a shortest killing continuation all the intermediate nonempty subsets have size at most k: they are images of q under longer words, so (1) applies. No nonempty subset can repeat along this continuation. If it did, deleting the factor between equal subsets would preserve the final empty

image and shorten the continuation. Counting the nonempty subsets before the final empty image shows that the continuation has length at most D.

List the original states $q_1, \dots, q_n$. Starting with the empty prefix, successively append a shortest killing continuation for the current image of q_i, skipping it if that image is already empty. A dead row remains dead under extensions, so after at most n continuations the global image is empty. The resulting mortal word has length at most nD.

Finally, order Q. Encode each nonempty subset of size at most k by listing its elements increasingly and padding the list to length k with copies of its maximum. This is an injection into Q^k, because the set of entries recovers the subset. Thus $D \le n^k$, giving $\mu(\mathcal{A}) \le n^{k+1}$. Together with the lower bound, this completes Theorem 1.

Scope and remaining extremal questions. The bounds leave room to improve constants and determine exact extremal thresholds. The distinction between endpoints and path multiplicity is visible in the construction: there are k different b^2-paths from p_0 to itself, and k^r such b^{2r}-paths. Their endpoints merge, in agreement with Lemma 3.

Computational checks. Exact finite checks regenerated the transition relations, exhausted the reachable global subset graphs and all singleton subset closures in small instances, and tested the potential identity and the constructive strategy. A separately implemented verifier covered 22 instances, including repeated and noncoprime clock lengths. These finite checks test implementation and boundary cases and complement the all-parameter proof above.

AI methodology. OpenAI Codex contributed substantially to all sections: developing the construction and proofs, comparing primary literature, implementing exact set and integer-bitmask checks, and drafting the exposition and LaTeX. Separate Codex contexts checked the argument and finite computations. Responsibility for any submission and its content remains with the named author.

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